CUHK-Shenzhen
Course Outline

1. **Course Identity**

A. **Course as listed in CUHK-Shenzhen**

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	FMA4200
Course title (English)	Financial Data Analysis
Course title (Chinese)	金融数据分析
Units	3
Description (English)	This course introduces several important tools for the analysis of financial data, including multivariate analysis, regression analysis, time series analysis, stochastic processes, principle component analysis, etc, with applications to risk management, derivatives pricing, portfolio management, etc.
Description (Chinese)	本课程主要介绍一些重要的金融数据分析方法，包括多元分析，回归分析，时间序列分析，主成分分析等；并基于这些方法解决一些重要的金融问题，包括风险管理，衍生品定价，投资组合管理等。

B. **Corresponding course in CUHK**

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in SZ maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	ECON3140
Course title (English)	Financial Data Analysis 金融数据分析
Course title (Chinese)	金融数据分析
Units	3
Description (English)	This course provides an introduction to the modern econometric techniques used in financial time series. Economic theory and statistical analysis are used to formulate and test financial models. Applications include the statistical characteristics and forecasting of stock prices, derivatives, interest rates and exchange rates.
Description (Chinese)	本课程介绍金融时间序列分析的现代计量技术。经济理论与统计分析被用来构造和检验金融模型。应用包括股票价格的统计特征和预测，衍生品，利率与汇率。

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1. Prerequisites

Having taken courses linear algebra, calculus, and probability theory.

A. Co-requisites

None.

2. Learning Outcomes

Knowledge:

a) Students will understand the mechanics of financial markets and the important computational finance problems.
b) Student will understand stochastic models of finance.
c) Students will understand finite difference method and its applications in solving finance problems.
d) Students will understand Monte Carlo method and its applications in solving finance problems.

Skills

Generic:

e) Students will be able to solve derivatives pricing problems with stochastic models and numerical methods.
f) Students will be able to evaluate and manage financial risk with stochastic models.

Valued/Attitude:

g) Students will have awareness of the power of stochastic models and numerical methods in solving real-world problems.

3. Course syllabus

Introduction to financial data analysis.
Statistical methods for risk management.
Volatility modeling and derivatives pricing.
Statistical methods for portfolio management.

4. **Assessment Scheme**

Component/ method	**% weight**
Assignments	30
Mid-term exam	30
Final project	40

5. **Descriptor**

Grade	**Overall course**
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

6. **Declaration on the use of AI tools**

Are students permitted to use AI tools (in particular GPT and generative AI tools) in the learning process and their assignment/course project? Please select one of the following approaches that will be adopted in this course.

☐ Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course.

■ Use only with explicit acknowledgement: Students are allowed or expected to use AI tools in their learning process and assignment/project reports. However, students must explicitly cite or acknowledge these resources and explain how they were used in their submitted reports. The course instructor will provide specific examples. Failure to follow this rule will be considered academic dishonesty.

☐ Free use of AI tools with no acknowledge: This course requires frequent use of AI tools, and students can use these tools throughout the learning process without the need to explicitly cite or acknowledge them. The instructor may specify which AI tools are allowed and set proper regulations if needed. Students who do not follow the regularations will be considered academic dishonesty.

7. **Feedback for evaluation**

A course questionnaire (CTE) was answered by each student after completing all teaching activities.
Reflections from teachers of more advanced courses.
Feedback from informal conversations over in the instructor's office and wechat/e-mail correspondence with students after class.

8. **Reading**

A. **Required**

Lecture notes prepared by the instructor.

B. **Recommended**

9. **Course components**

Activity	**Hours/week**
Lectures	3
Tutorial	1
Readings & Assignments	5-6

10. **Indicative teaching plan**

Week 1: Introduction to financial data analysis.
Week 2-5: Statistical methods for risk management.
Week 3-8: Volatility modeling and derivatives pricing.
Week 9-14: Statistical methods for portfolio management.